

Fraud Detection Project Update

Meeting with Head of Data Science

Duration: 20-25 minutes

Date: [Current Date]

1. PROJECT STATUS SUMMARY (3 mins)

What We've Completed

Access & Infrastructure (8 weeks total)

- Secured VM access, database setup, and DFS data permissions
- Successfully connected to IAR (transaction data) and MBAR (customer data)
- Initial data exploration completed

Key Data Understanding

- **Volume:** ~250M transactions/month, ~10M/day, ~300K/hour at peak
- **Timeframe:** January to July 2025 (7 months of data)
- **Fraud cases:** 40K transactions marked as fraud
- **Complaint volume:** ~6,000 complaints/day from ServiceNow CPS

Current Phase

We're in **data exploration and preparation** with 1-2 weeks remaining before model building begins.

2. KEY INSIGHTS FROM ANALYSIS (5 mins)

Pattern Discovery

1. **70% spike in fraud cases** in May 2025 (primarily JC App and Payment Gateway)
2. **1st of month concentration** - highest fraud complaints on monthly cycle
3. **Peak fraud window:** 1-7 PM (aligns with high transaction volume)
4. **Channel concentration:** 88% of fraud involves JC App, Payment Gateway, or Third-Party sites
5. **Transaction types:** 80% are P2P transfers, payments, and loan-related

Critical Finding - Payment Gateway Risk

- **27% fraud ratio** in Payment Gateway (down from 41% in July)
- Highest risk channel by significant margin
- Most transactions use MPIN authentication

Data Quality Challenges

1. **Complaint-based labeling only** - no automated fraud rules
 - 30-40 min delay between complaint and fraud marking
 - Labels are NEVER updated (even false positives stay marked)
2. **Extreme class imbalance:**
 - Fraud rate: 0.0023% (40K fraud / 1.75B transactions)
 - Only 3.2% of complaints result in fraud marking
3. **Duplicate entries** identified in fraud data (needs cleaning)

3. SCOPE DEFINITION (2 mins)

Phase 1 Focus (Current)

INCLUDED:

- Customer-to-Customer (C2C) transactions only
- Customer accounts: Levels 0, 1, 2
- Channels: Payment Gateway, JC App, USSD, Third-Party Apps
- Account statuses: NEW and ACTIVE only

EXCLUDED:

- Organizational/Parking accounts (different fraud patterns)
- Merchant accounts (Level 3)
- Dormant/Suspended accounts (cannot transact)

Rationale: Focusing on high-volume C2C segment for faster MVP development.

4. TECHNICAL CHALLENGES & GUIDANCE NEEDED (10 mins)

Challenge 1: Extreme Class Imbalance

The Problem:

- 0.0023% fraud rate (2.3 frauds per 100,000 transactions)
- 40K fraud cases vs 1.75B total transactions
- Standard sampling creates datasets too large or loses signal

Our Proposed Approaches:

Option A: Complaint-Centered Sampling



- ALL fraud cases: 40K
- Sample non-fraud complaints: 200K (from 1.26M complaints)
- Stratified random transactions: 1M
- Total: ~1.24M samples with ~3% fraud rate

Option B: Stratified Transaction Sampling



- ALL fraud cases: 40K
- Payment Gateway: 5% sample (high-risk)
- Peak hours (6-8 PM): 2% sample
- 1st of month: 10% sample
- Regular transactions: 0.1% sample
- Total: ~2M samples with ~2% fraud rate

Option C: Hybrid Matched Controls



- ALL fraud cases: 40K
- Matched controls: For each fraud, find similar non-fraud (1:25 ratio)
- Complaint-based non-fraud: 200K
- Random sample: 500K
- Total: ~1.74M samples with ~2.3% fraud rate

? Question 1: Which sampling strategy would you recommend given the extreme imbalance? Should we consider ensemble approaches with different sampling strategies?

Challenge 2: Label Quality Issues

The Problem:

- Fraud marked ONLY through customer complaints (no ground truth)
- Labels never corrected (false positives remain marked as fraud)
- We're predicting "reported and confirmed complaints" not actual fraud

Implications:

- Unknown volume of unreported fraud (silent victims)
- False positives in training data (mutual resolutions still marked as fraud)
- Model will learn to predict "reportable fraud" not "all fraud"

Our Considerations:

- Should we treat this as a semi-supervised learning problem?
- Consider label confidence weighting based on:
 - Time to complaint (faster = more confident)
 - Repeat fraudster (fraudster with multiple victims = more confident)
 - Account suspension status (if unsuspended later = less confident)

? Question 2: How should we handle label noise? Should we:

- Accept it and document the limitation?
 - Build label confidence scores into the model?
 - Use anomaly detection approaches alongside supervised learning?
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Challenge 3: Model Architecture Selection

The Context:

- Two potential prediction targets:
 - Transaction-level: Will this transaction be fraud? (0.0023% base rate)
 - Complaint-level: Is this complaint actual fraud? (3.2% conversion rate)

Proposed Approaches:

Option A: Single-Stage Model



Input: Transaction features at time T

Output: Fraud probability

Focus: Real-time transaction scoring

Option B: Two-Stage Pipeline



Stage 1: Complaint Risk Model

- Input: Transaction features
- Output: Complaint probability
- Use: Early warning system

Stage 2: Fraud Confirmation Model

- Input: Transaction + complaint features
- Output: Fraud probability
- Use: Complaint triage

Option C: Ensemble with Channel-Specific Models



- Payment Gateway model (27% fraud ratio - different patterns)
- JC App model
- General transaction model
- Combine predictions with meta-learner

? Question 3: Given our data characteristics and business needs, which architecture would you prioritize for the MVP? Should we build for real-time transaction blocking or post-transaction complaint triage?

Challenge 4: Feature Engineering Strategy

Current Plan:

Temporal Features:

- Hour of day, day of week, day of month
- Time since account creation
- Transaction velocity (rolling windows: 1hr, 24hr, 7d, 30d)

User Aggregation Features:

- Historical transaction patterns
- Average transaction amount
- Channel usage distribution
- Complaint history

Transaction Features:

- Amount, channel, transaction type
- Deviation from user's normal behavior
- Account level, status, age

Network Features (if time permits):

- Recipient account risk score
- Connected component analysis
- Graph-based fraud propagation

? Question 4: Given our 2-3 week timeline for model building:

- Should we prioritize breadth (many feature types) or depth (sophisticated features in fewer categories)?
 - Are there specific feature engineering patterns that work well for extreme imbalance?
 - Should we invest time in graph/network features or keep it simpler for MVP?
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Challenge 5: Model Evaluation Strategy

The Problem:

- Standard accuracy is meaningless (99.9977% by always predicting non-fraud)
- Precision/Recall trade-offs critical but what's the business threshold?
- Can't evaluate on unreported fraud

Proposed Metrics:

Primary Metrics:

- Precision @ Top K (for operational deployment)
- Recall @ Fixed Precision (business constraint)
- F-beta score (weight recall higher than precision?)

Secondary Metrics:

- Fraud detection rate within top 1% scored transactions
- Channel-specific performance (especially Payment Gateway)
- Temporal stability (performance over time)

Business Metrics:

- Estimated fraud loss prevented
- False positive rate impact on customer experience
- Complaint handling efficiency improvement

? Question 5:

- What should be our primary success metric for the MVP?
- What precision/recall trade-off is acceptable for the business?
- Should we set different thresholds for different channels (e.g., stricter for Payment Gateway)?

Challenge 6: Real-time Deployment Considerations

Questions to Clarify:

- Will this be real-time transaction blocking or batch processing?
- If real-time, what's the acceptable latency? (<100ms? <1s?)
- What features can we access in real-time vs batch?
- How do we handle model/data drift in production?

? **Question 6:** Should we design the MVP with real-time constraints in mind, or focus on batch processing first and optimize for latency later?

5. RECOMMENDED APPROACH FROM OUR SIDE (2 mins)

Based on our analysis, we're leaning toward:

Sampling: Hybrid approach (Option C) with matched controls + complaint data + stratified sample

Model Architecture: Two-stage pipeline (Option B)

- Stage 1 for early warning on all transactions
- Stage 2 for complaint triage with higher accuracy

Feature Strategy: Focus on user aggregation + temporal patterns + channel-specific features (skip graph features for MVP)

Evaluation: Precision @ Top 1% as primary metric, targeting 50%+ precision while maintaining 30%+ recall

MVP Timeline:

- Week 1: Data preparation with chosen sampling
- Week 2: Model training and validation
- Week 3: Evaluation, documentation, and handoff

But we'd love your input on whether this aligns with business priorities and technical best practices.

6. ADDITIONAL QUESTIONS (3 mins)

Business Context

1. **What's the estimated monetary loss per fraud case?**
 - This helps us calculate ROI and prioritize detection vs false positive trade-offs
2. **What's more valuable: catching fraud early (real-time) or triaging complaints efficiently (post-transaction)?**
 - Guides our model architecture decision
3. **Is there a target fraud detection rate?**

- E.g., "We want to catch 50% of all fraud" vs "We want to reduce fraud loss by 30%"

Technical Constraints

4. **What's the deployment environment?**
 - Cloud vs on-premise? Latency requirements? Scalability needs?
5. **Who will maintain the model post-deployment?**
 - Affects complexity decisions and documentation needs
6. **Are there existing fraud rules/systems we need to integrate with?**
 - Should our model complement or replace current systems?

Data Access

7. **Can we access complaint text/descriptions?**
 - NLP features could improve fraud vs non-fraud complaint classification
8. **Is there any feedback loop data?**
 - Cases where fraud was detected but later reversed?
 - Mutual resolution records?

Phase 2 Considerations

9. **After C2C fraud, what's the next priority?**
 - Merchant fraud? Organizational accounts? Other fraud types?
10. **What's the timeline expectation for production deployment?**
 - Affects how much we optimize the MVP vs quick iteration

7. NEXT STEPS BASED ON GUIDANCE (1 min)

After this meeting, we'll:

1. Implement the agreed sampling strategy
2. Begin feature engineering with prioritized feature groups
3. Set up evaluation framework with confirmed metrics
4. Build MVP model with chosen architecture
5. Document findings and prepare for deployment discussion

Expected deliverables in 3 weeks:

- Trained fraud detection model
 - Performance evaluation report
 - Feature importance analysis
 - Deployment recommendations
 - Documentation for handoff
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APPENDIX: Quick Reference Numbers

Metric	Value
Total Transactions (7 months)	1.75B
Monthly Transaction Volume	~250M
Daily Transaction Volume	~10M
Peak Hourly Volume	~300K
Total Fraud Cases	40K
Fraud Rate	0.0023%
Daily Complaints	~6,000
Complaint → Fraud Conversion	3.2%
Payment Gateway Fraud Ratio	27%
Top 3 Channels (fraud %)	88%
P2P/Payment Transactions (fraud %)	80%

Thank you! Looking forward to your guidance.